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The Cost of Mismatch

Noise Amplification in Zeroth-Order Reinforcement Learning Scales as \(1/\mu^2\); Matched-Data BOLT Cancels It

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Zeroth-order reinforcement learning for large language models suffers from a fundamental noise amplification effect in the mismatched-data setting: gradient variance scales as \(1/\mu^2\), creating an irreducible noise floor as the perturbation scale \(\mu\) shrinks. The authors identify the matched-data BOLT estimator as the antidote, derive ZO-BOLT, and demonstrate 3–5\(\times\) lower gradient variance and 40% faster convergence on GSM8K and HumanEval.

AT A GLANCE

ZO RL for LLMs is unstable in practice; the reason is not understood theoretically.

ZO methods are the only approach that enables policy-gradient RL without backpropagation, keeping memory at inference cost; instability blocks their practical use.

The authors prove a \(1/\mu^2\) variance lower bound for mismatched-data ZO RL and derive ZO-BOLT, which shares rollout randomness between both perturbation evaluations to cancel the dominant noise term.

ZO-BOLT reduces gradient variance by 3–5\(\times\) and reduces forward passes to target accuracy by 40% on GSM8K and HumanEval.

Gradient variance measurements; convergence curves on mathematical reasoning (GSM8K, MATH) and code generation (HumanEval).

The Big Picture

Policy gradient RL requires differentiating through a reward signal that is typically non-differentiable (discrete text actions, binary correctness signals). Zeroth-order methods estimate the gradient by measuring reward differences between two parameter-perturbed policies, avoiding backpropagation entirely. This makes ZO RL attractive for fine-tuning LLMs with RL, since the memory footprint equals that of inference.

Despite this appeal, practitioners have found ZO RL for LLMs to be unstable—high variance, slow convergence, sensitive to hyperparameters. In contrast, ZO supervised fine-tuning methods such as MeZO work reliably. This discrepancy motivates the central question of the paper: is there a fundamental difference between ZO SFT and ZO RL that explains the instability?

Why Existing Approaches Fall Short

Standard ZO RL uses the two-point estimator: $$\hat{g} = \frac{r(\theta + \mu z) - r(\theta - \mu z)}{2\mu} \cdot z$$ where the two reward evaluations use independently sampled rollouts. The mismatch between these rollouts is the fundamental problem.

Smaller perturbation scale μ reduces the bias of the estimator but, as the paper shows, the variance grows proportionally to $1/\mu^2$—the two effects cannot be balanced simultaneously.

More rollouts per update reduces variance linearly but does not address the $1/\mu^2$ structure, requiring a quadratic increase in rollouts to halve the variance floor.

Core Mechanism: The $1/\mu^2$ Lower Bound

Let $r(\theta)$ be the stochastic reward with noise variance σ^2_env. The paper shows that the variance of the standard ZO RL estimator decomposes as: $$\mathrm{Var}\!\left(\hat{g}\right) = \frac{\sigma^2_\text{env}}{2\mu^2} + O\!\left(\frac{\sigma^4_\text{env}}{\mu^4}\right) + \text{signal terms}$$

The leading term $\sigma^2_\text{env}/(2\mu^2)$ grows without bound as $\mu \to 0$, independent of sample size. This is the *mismatch penalty*: it arises specifically because the two evaluations $r(\theta + \mu z)$ and $r(\theta - \mu z)$ use different rollout seeds, so their noise does not cancel in the difference.

The BOLT Solution

Boltzmann Targeted SFT (BOLT) in the matched-data setting shares the same rollout seed for both evaluations. Under matched randomness: $$\tilde{g}_\text{BOLT} = \frac{r_s(\theta + \mu z) - r_s(\theta - \mu z)}{2\mu} \cdot z$$ where both evaluations use rollout seed s. The noise now cancels: $$r_s(\theta + \mu z) - r_s(\theta - \mu z) = \nabla_\theta r(\theta)^\top (2\mu z) + O(\mu^3)$$ with no leading noise term. The variance reduces to: $$\mathrm{Var}\!\left(\tilde{g}_\text{BOLT}\right) = O(1)$$ independent of μ, eliminating the $1/\mu^2$ amplification entirely.

ZO-BOLT Algorithm

ZO-BOLT instantiates matched-data perturbation for LLM RL fine-tuning:

1. Sample a random direction z in parameter space.
2. Fix a rollout seed s.

3. Evaluate the model under $\theta + \mu z$ with seed s to obtain reward r^+.
4. Evaluate the model under $\theta - \mu z$ with the same seed s to obtain reward r^-.
5. Compute gradient estimate $\hat{g} = (r^+ - r^-)/(2\mu) \cdot z$.
6. Update: $\theta \leftarrow \theta - \eta \hat{g}$.

The matched-seed constraint adds zero computational overhead—it requires only fixing the random seed before each rollout pair.

What the Guarantee Says

Theorem 1 (informal): For any ZO estimator in the mismatched-data setting with environment noise $\sigma^2_{\text{env}} > 0$, the gradient variance is at least $\sigma^2_{\text{env}}/(2\mu^2)$ for perturbation scale μ. No mismatched estimator can circumvent this bound.

Theorem 2 (informal): The ZO-BOLT matched-data estimator achieves $O(1)$ variance, independent of μ. Crucially, this requires only that $r_s(\cdot)$ is deterministic given the seed, which holds for any pseudorandom rollout procedure.

The bounds are non-asymptotic and hold for finite d and $\mu > 0$.

Experimental Findings

- **Gradient variance (GSM8K):** ZO-BOLT reduces gradient variance by $3.2\times$ over standard ZO RL at $\mu = 10^{-3}$, and $5.1\times$ at $\mu = 10^{-4}$.
- **Convergence (GSM8K):** ZO-BOLT reaches 70% accuracy with 40% fewer forward passes than standard ZO RL.
- **Convergence (HumanEval):** ZO-BOLT reaches 50% Pass@1 with 35% fewer forward passes.
- **MATH:** ZO-BOLT final accuracy improves by 4.3% over standard ZO RL with the same forward-pass budget.

Ablations and Interpretation

Effect of μ on ZO-BOLT: Unlike standard ZO RL, ZO-BOLT performance is insensitive to μ across the range 10^{-4}–10^{-2}, confirming that the mismatch penalty has been eliminated.

Seed granularity: The paper evaluates fixing the seed per rollout pair (default) vs. per batch. Per-pair seeding gives lower variance, as expected from the theory.

Comparison to first-order RL: ZO-BOLT is \(12\times\) slower to converge than GRPO (a first-order RL method) on GSM8K, but uses \(8\times\) less memory—a favorable trade-off when GPU memory is the bottleneck.

Reference

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